1. **Aim of the practical:** Process and job scheduling commands like ps, stop, kill, fb, bgcrontab, cron etc.

2. **Objective: To acquire foundational skills in Linux process management and automation by utilizing cron jobs to schedule tasks, managing processes, and editing script files effectively.**

3. Hardware and Software Requirements:

a) **Disk Drive**: Physical disk (HDD/SSD) with enough space.
   Computer System: Host system running virtualization software.
b) **Operating System**: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step 1: Check Running Processes

```
nikhilyadav@fedora:~$ ps
    PID TTY          TIME CMD
   2906 pts/0    00:00:00 bash
   3270 pts/0    00:00:00 ps
```

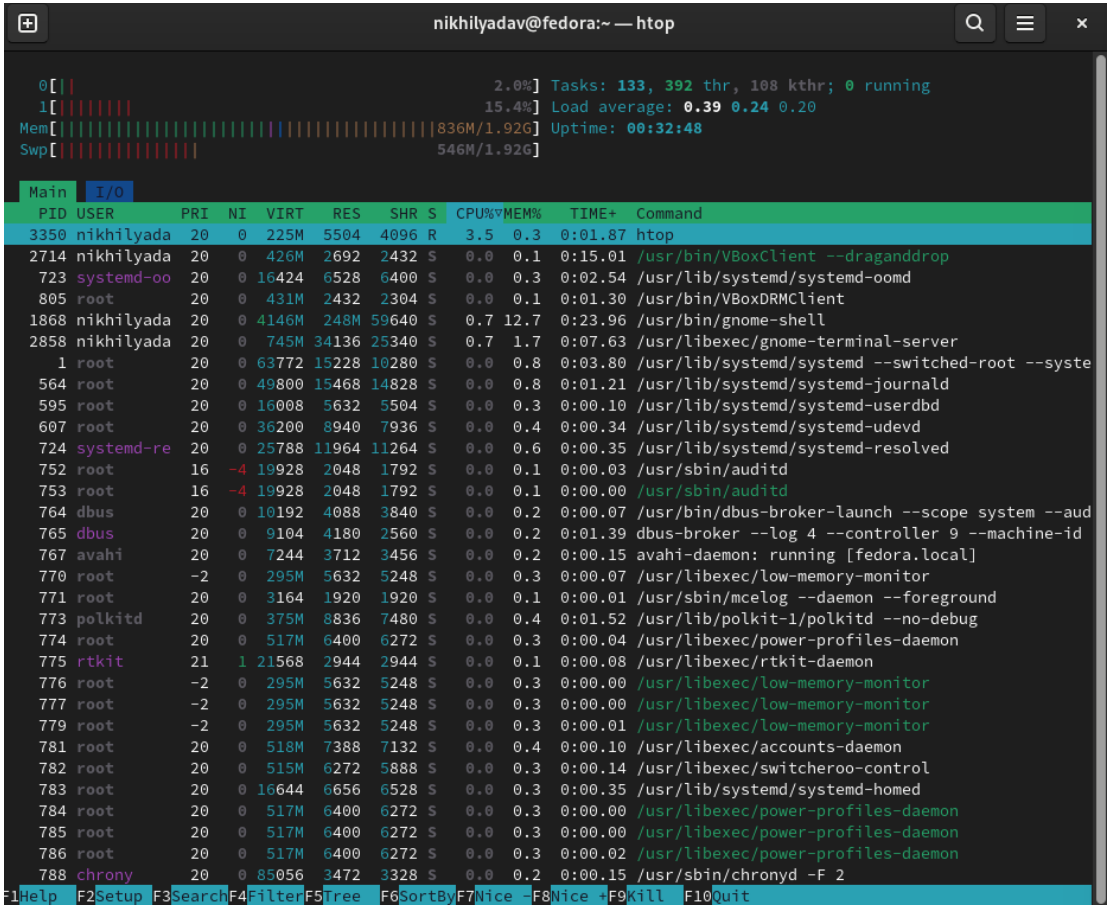
Step 2: View Detailed Process Information

```
nikhilyadav@fedora:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.7  63772 15356 ?        Ss   22:16   0:03 /usr/lib/syst
root           2  0.0  0.0      0     0 ?        S    22:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    22:16   0:00 [pool_workque
root           4  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-rc
root           5  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-rc
root           6  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-sl
root           7  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-ne
root           8  0.0  0.0      0     0 ?        I    22:16   0:01 [kworker/0:0-
root          13  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-mm
root          14  0.0  0.0      0     0 ?        I    22:16   0:00 [rcu_tasks_kt
root          15  0.0  0.0      0     0 ?        I    22:16   0:00 [rcu_tasks_ru
root          16  0.0  0.0      0     0 ?        I    22:16   0:00 [rcu_tasks_tr
root          17  0.0  0.0      0     0 ?        S    22:16   0:00 [ksoftirqd/0]
root          18  0.0  0.0      0     0 ?        I    22:16   0:01 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    22:16   0:00 [migration/0]
root          20  0.0  0.0      0     0 ?        S    22:16   0:00 [idle_inject/
root          21  0.0  0.0      0     0 ?        S    22:16   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    22:16   0:00 [cpuhp/1]
root          23  0.0  0.0      0     0 ?        S    22:16   0:00 [idle_inject/
root          24  0.0  0.0      0     0 ?        S    22:16   0:00 [migration/1]
root          25  0.1  0.0      0     0 ?        S    22:16   0:01 [ksoftirqd/1]
root          26  0.1  0.0      0     0 ?        I    22:16   0:02 [kworker/1:0-
root          28  0.1  0.0      0     0 ?        I    22:16   0:01 [kworker/u5:0
root          30  0.0  0.0      0     0 ?        S    22:16   0:00 [kdevtmpfs]
root          31  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-in
root          32  0.1  0.0      0     0 ?        I    22:16   0:02 [kworker/u6:1
root          33  0.0  0.0      0     0 ?        S    22:16   0:00 [kauditd]
root          35  0.0  0.0      0     0 ?        S    22:16   0:00 [oom_reaper]
root          36  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-wr
root          37  0.0  0.0      0     0 ?        S    22:16   0:00 [kcompactd0]
root          38  0.0  0.0      0     0 ?        SN   22:16   0:00 [ksmd]
root          39  0.0  0.0      0     0 ?        SN   22:16   0:00 [khugepaged]
root          40  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-cr
root          41  0.0  0.0      0     0 ?        I<   22:16   0:00 [kworker/R-ki
```

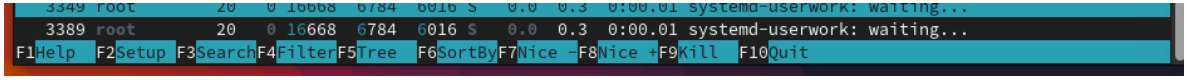
We installed the htop using sudo dnf install htop

```
nikhilyadav@fedora:~$ sudo dnf install htop
[sudo] password for nikhilyadav:
Last metadata expiration check: 0:15:30 ago on Sun 20 Oct 2024 10:28:41 PM IST.
Dependencies resolved.
================================================================================
 Package            Architecture   Version                Repository       Size
================================================================================
Installing:
 htop               x86_64         3.3.0-3.fc40           fedora          195 k
Installing dependencies:
 hwloc-libs         x86_64         2.11.1-1.fc40          updates         2.1 M

Transaction Summary
================================================================================
Install  2 Packages

Total download size: 2.3 M
Installed size: 3.3 M
Is this ok [y/N]: y
Downloading Packages:
(1/2): htop-3.3.0-3.fc40.x86_64.rpm              74 kB/s | 195 kB     00:02
(2/2): hwloc-libs-2.11.1-1.fc40.x86_64.rpm      372 kB/s | 2.1 MB     00:05
--------------------------------------------------------------------------------
Total                                           318 kB/s | 2.3 MB     00:07
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : hwloc-libs-2.11.1-1.fc40.x86_64                        1/2
  Installing       : htop-3.3.0-3.fc40.x86_64                               2/2
  Running scriptlet: htop-3.3.0-3.fc40.x86_64                               2/2

Installed:
  htop-3.3.0-3.fc40.x86_64            hwloc-libs-2.11.1-1.fc40.x86_64

Complete!
```

We used Htop command to see all the running tasks.



Opened a new terminal and killed it using F9 or we can kill using kill <process id>



Now installed Crontab

```
nikhilyadav@fedora:~$ crontab -e
bash: crontab: command not found...
Install package 'cronie' to provide command 'crontab'? [N/y] y


 * Waiting in queue...
 * Loading list of packages....
The following packages have to be installed:
 cronie-1.7.2-1.fc40.x86_64     Cron daemon for executing programs at set times
 cronie-anacron-1.7.2-1.fc40.x86_64     Utility for running regular jobs
 crontabs-1.11^20190603git9e74f2d-3.fc40.noarch Root crontab files used to schedule the execution of programs
Proceed with changes? [N/y] y


 * Waiting in queue...
 * Waiting for authentication...
 * Waiting in queue...
 * Downloading packages...
 * Requesting data...
 * Testing changes...
 * Installing packages...
no crontab for nikhilyadav - using an empty one
Standard input is not a terminal
crontab: "/usr/bin/nano" exited with status 1
```

Now Made new executable file for making crontab use

```
nikhilyadav@fedora:~$ touch ~/script.sh
nikhilyadav@fedora:~$ vim ~/script.sh
nikhilyadav@fedora:~$ chmod +x ~/script.sh
```

Verifying the crontab

```
nikhilyadav@fedora:~$ crontab -l
no crontab for nikhilyadav
```

- At 5:00 PM every day, the cron daemon (the background service that runs cron jobs) will execute the command . This runs the script you created, which will print "Hello!" to the console.

6. Learning outcomes (What I have learned):

- **Understanding Cron Jobs**: Learned how to schedule automated tasks using cron jobs in Linux.
- **File Manipulation**: Gained experience in creating and editing script files using vim.
- **Process Management**: Practiced managing processes using commands like ps, htop, and kill.
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Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature